

BOOK ERRATA –
Foundational and Applied Statistics for Biologists using R
2nd printing, 1st edition
updated 4/20/2026

Chapter 1

1. Page 19, Q 10. *reductio~~n~~ ad absurdum*

Chapter 2

1. Page 31. The upper bound to the probability of an intersection ~~occurs under independence will equal~~
the probability of the event with the smallest marginal probability: $\min(P(A), P(B), \dots, P(k))$.
2. Page 47, Q 31d. Given results in (~~b~~e) and (~~c~~d)

Chapter 3

1. Page 70, Example 3.7. Recently, ~~88~~120 goats were randomly sampled from the GYE...
2. Pg. 93. The mean of a geometric distribution is $(1 - \pi)/\pi$, not $1/\pi$ (Thanks to Dr. Brian Parker, New York University)
3. Pg. 98, Q. 13.
... for negative binomial dispersion parameter ~~m~~k and mean ~~k~~m.
4. Pg. 98, Q. 13 part e.
`Lambda.hat <- mean(Leaf.obs) mean(rep(Mites.per.leaf, Leaf.obs))`

Chapter 4

1. Pg. 144. Q. 8. In the table, X should be replaced with x .
2. Pg. 146 Q. 29. “Let $X \sim N(2,4)$, ~~and let~~ $Y \sim N(3,2)$, and assume that X and Y are independent.”
3. Pg. 146. Q. 30. To the end of the paragraph describing the problem, the following statement should be added: “Assume J and B are independent.”

Chapter 5

1. Pg. 167. In Grand Teton National Park ~~in~~
2. Pg. 193. Q. 1. To the end of the paragraph describing the problem, the following statement should be added: “Assume J and B are independent.”

- Pg. 194. Q. 7. To the end of the sentence describing the problem, the following statement should be added: “Assume random variables are independent, and that H_0 is true for (d) and (e).”

Chapter 6

- Page 216. An error occurs in the worked example for MSE and t^* .
The example *currently* reads:

$$t^* = \frac{(0.114 - 0.099) - 0}{0.0171\sqrt{2/34}} = 2.4607$$

It should be changed to read:

$$t^* = \frac{(0.114 - 0.099) - 0}{0.0171\sqrt{3/34}} = 2.4607$$

- Page 223. She wants to use $\alpha = 0.05$ and $1 - \beta = 0.8$.
- Page 223.

Replace:

$$n = \frac{(1.645 - 0.842)^2 100}{(-5 - 0)^2} = 24.7$$

With:

$$n = \frac{(1.645 + 0.842)^2 100}{(-5 - 0)^2} = 24.7$$

- Page 223. Given $\alpha = 0.05$. $1 - \beta = 0.8, \dots$
- Page 234.
In a lower tailed test, the test statistic W^* will also be ~~W_2~~ W_1 .
- Page 241, Question 4. Last sentence in the Question 4 introduction should have the following corrections.
“With this in mind, I take a random sample of 20 female freshmen honors students and find that the mean height is 61 inches.”
- Page 243, Question 9. By typing `book.mneumenu()`

Chapter 7

- Page 261. ...area is ~~$\{(61.59, 94.34)\}$~~
- Page 281. ...~~block~~randomize what you can randomize-block what you cannot
- Page 291. Question 12. A sample represents a 5 km^2 area.
- Page 291. Question 12b and 12c. Change the order of these questions.
- Page 291. Question 12d. Replace the symbol $S_{\bar{Y}_{str}}$ with $\hat{\sigma}_{\bar{Y}_{str}}$

6. Page 291. Question 12e. Replace the symbol $S_{\hat{t}}$ with $\hat{\sigma}_{\hat{t}}$

Chapter 8

1. Page 306. Missing right parenthesis in code:

```
with(crab.weight, cor.test(gill.wt, body.wt, method = "pearson"))
```

Chapter 9

1. Page 345. Text preceding last chunk on page. Thus we are 95% confident that the mean of Y given $X = 3080$ will lie in the interval (478.35, 510.32) ~~(541.12, 573.08)~~, and we are 95% confident that a future observation of Y given $X = 3080$ will lie in the interval (4344.40, 554.26) ~~(497.17, 617.03)~~.
2. Page 346. Eq. 9.39. For notational consistency, replace TSS with SSTO in denominators of ratios. That is, replace:

$$R^2 = 1 - \frac{SSE}{TSS} = \frac{SSR}{TSS}$$

with

$$R^2 = 1 - \frac{SSE}{SSTO} = \frac{SSR}{SSTO}$$

3. Page 347. Eq. 9.40 is incorrect. Replace:

$$R_{adj}^2 = 1 - \frac{SSE}{SSTO} \times \frac{n-1}{n-p}$$

with

$$\begin{aligned} R_{adj}^2 &= 1 - \left(\left(1 - \frac{SSR}{SSTO} \right) \left(\frac{n-1}{n-p} \right) \right) \\ &= 1 - \left(\left(1 - R^2 \right) \left(\frac{n-1}{n-p} \right) \right) \end{aligned}$$

4. Page 358. Text above Eq. 9.54. The reciprocal of the denominator of Equation 9.53 is the k th VIF. That is
5. Page 358. Text below Eq. 9.53. Note that as R_k^2 increases, ~~$1/(1 - R_k^2)$~~ approaches 0, ...
6. Page 368. Text above Example 9.21. ...then the two-tailed P -value is calculated as $2P(T \geq |t^*|)$...
7. Page 389. Eq. 9.86. Use of λ is for log-likelihood is confusing. Replace occurrences of λ with ℓ .
8. Page 392. Last sentence of section 9.20.5.2

The *sensitivity* (i.e. *true positive rate*) is the number of true positives divided by the number of ~~observed~~ true positives and false negatives: $14/15 = 0.93$. The *specificity* (i.e., *true negative rate*) is the number of true negatives divided by the number of ~~observed-true~~ negatives and false positives: $6/9 = 0.67$ (also see Example 2.11).

9. Page 403. 1st sentence of section 9.22. We calculate estimates of parameters for a ...

10. Page 417. Question 10.

bc. Adjust the slider widgets, or simply click *Refresh* repeatedly (> 30 times). Are *MSE* and *MSR* consistently greatly-greater than or less than $E(MSE)$ and $E(MSR)$. Why?

Chapter 10

1. Page 447. Eq. 30. Replace σ_{2A} with σ_A^2 .

2. Page 497. Question 1. “Described in Exercise-Example 10.2”.

3. Page 500. Question 13. To allow adequate replication use the following data:

```
scores <- data.frame(APscores = c(4.6, 4.8, 4, 3.4, 3.4, 2.1, 4.1, 3.9,
4, 4.1, 3.8, 1.2, 4.5, 4.9, 4.3, 3.5, 3.6, 3.2, 3.9, 4.3, 3.8, 4.2, 4.4,
2.4), school = factor(rep(c(1,2), each = 6)), town = c("Burley",
"Pocatello", "Boise"), instructor = factor(rep(c(rep(1, 3), rep(2, 3)),
2)), class = factor(rep(c(1,2), each=12)))
```

4. Page 500. Question 16a. Analyze the data correctly (~~with block in hybrid as a random effect~~) using aov.

4. Page 500. Question 16c. Reanalyze the data, and repeat the hypothesis tests using lmer. Define block and hybrid as random effects.

Appendices

1. Page 549, A.4 Set Theory and Probability ~~$P(A)$~~ $1 - 1 - P(A) = P(A')$

Notes for next edition:

Check additions/revisions to Chapter exercises in Canvas/Moodle.